MADA: A Multi-Agent AI System for Autonomous Database Administration

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Abstract

Database administration — indexing, configuration, query tuning, incident diagnosis and security monitoring — is still largely manual, and the AI tools that automate it each cover one duty and act on predictions rather than measurements. This project proposes MADA (Multi-Agent Database Administrator), a multi-agent system for autonomous PostgreSQL administration. Six specialist agents (monitor, index, knob, query, diagnosis and security) will read shared telemetry, test candidate actions on a sandbox clone of the database, and submit structured proposals to a coordinator that resolves conflicts over shared resources and to a safety gate that admits only changes with measured gain, keeps irreversible actions for human approval and rolls back regressions. The system will be built as a working prototype on an e-commerce database under a live multi-role workload and evaluated on query latency, throughput, fault detection and repair, attack detection, and the effect of removing the safety gate.

Introduction

Almost every modern application depends on a relational database, and that database stays fast and safe only through continuous expert work: choosing indexes, setting memory and planner parameters, rewriting slow SQL, diagnosing incidents and watching for misuse. Pavlo et al. argued that a database management system should take on much of this work itself — a “self-driving” system that forecasts its own workload and plans its own changes [1]. In the years since, research on AI for databases has produced learned components for nearly every one of these duties [2]: knob tuners that learn configurations from earlier tuning sessions [3], cloud services that index millions of databases automatically [4], learned optimisers that steer query plans with hints [6], and diagnosis tools that explain anomalies from performance metrics [7].

These advances arrive as separate tools, each owning one duty and each trusting its own predictions, and two problems follow. First, the duties interact. A new index changes query plans, memory use and the cost of every write, so a knob tuner, an index advisor and a query rewriter running side by side can undo one another's work. Second, a prediction is not an outcome: Ding et al. showed that optimiser cost estimates frequently recommend indexes that make real queries slower [5]. Security sits further apart still — detectors that profile each role's normal access [9] usually run outside the tuning loop, so nothing decides whether a performance change or a security response should come first.

Large language models add a new option and a new risk. An LLM agent can read documentation and diagnose incidents [8], and several agents can divide a task between them through conversation [10]. But an agent that holds database credentials can also run a wrong statement directly on production, and research prototypes rarely put a verified safety layer between an agent's proposal and the live system.

This project proposes **MADA (Multi-Agent Database Administrator)**. Six specialist agents — monitor, index, knob, query, diagnosis and security — share one view of a PostgreSQL database, test every idea on a sandbox clone, and send structured proposals to a coordinator and a safety gate. The coordinator resolves conflicts over shared resources and orders actions by priority (security, availability, correctness, performance). The gate applies only changes backed by measured evidence, keeps

irreversible actions for a human, and rolls back any change that makes the system slower. A working prototype will be built on PostgreSQL 16 and evaluated on an e-commerce database under a live multi-role workload, with injected faults and an ablation study of the safety gate.

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